Product Requirement Document (PRD)

1. Document Information

- **Project Title:** AI-Powered Application Validation, Document Authentication, and Regulatory Compliance Verification Platform
- **Project ID:** SIH26261 (Smart India Hackathon)
- **Organization:** All India Council for Technical Education (AICTE), Regulation Bureau
- **Theme:** Smart Automation
- **Target Audience:** AICTE Officials, Technical Institutions, System Administrators

2. Product Overview

2.1 Purpose

The AICTE processes applications from technical institutions for Extension of Approval (EoA) and other changes based on the Approval Process Handbook. Currently, manual scrutiny of high-volume, varied applications and supporting documents is time-consuming. This leads to delays caused by missing/expired documents, inconsistencies, suspected alterations, and difficulties in confirming authenticity with original issuers.

This product aims to build a secure, AI-enabled decision-support platform integrated with the AICTE portal. It will perform preliminary validation, authenticate documents, and assist in checking regulatory compliance early in the process.

2.2 Goals & Objectives

- **Identify Deficiencies:** Flag missing, incomplete, expired, or inconsistent information/documents at submission.
- **Reduce Redundancy:** Utilize existing institutional records in authorized AICTE systems to minimize repeated manual checks.
- **Detect Anomalies:** Identify potential document alterations, duplication, and inconsistencies for further verification.
- **Automate Verification:** Authenticate documents via issuing-authority APIs, portals, digital repositories, QR codes, or digital signatures where available.
- **Infrastructure Compliance Assistance:** Use configurable rules (and computer vision where feasible) to check infrastructure evidence (photos, floor plans) against requirements, flagging issues for manual review or inspection.
- **Ensure Transparency:** Provide traceable, explainable validation results to support consistent regulatory processing.

2.3 Scope

- **In Scope:** Integration with AICTE application portal/data sources; OCR and data extraction; cross-document consistency checks; anomaly detection; third-party issuing authority verification workflows; compliance assistance module; validation dashboard with RBAC and audit trails.
- **Out of Scope:** Final regulatory decision-making (the system is strictly decision-support; AICTE authorities retain final approval). Bypassing access controls for restricted external databases.

3. User Personas

Persona	Role	Primary Objectives
AICTE Official	Reviewer / Approver	Reviews AI-generated flags, anomalies, and verification statuses to make final regulatory decisions.
Institution Applicant	Submitter	Submits applications and supporting documents; receives real-time feedback on missing or incomplete items.
System Administrator	System Configurator	Configures regulatory rules and checklists based on the Approval Process Handbook; manages RBAC.

4. Key Features & Functional Requirements

4.1 Intelligent Application & Document Validation

- **Category Identification:** Automatically identify the application category and apply the relevant checklist.
- **Record Retrieval:** Fetch existing records from AICTE systems (affidavits, accreditation, previous EoA, faculty records, etc.).
- **OCR & Data Extraction:** Extract text and metadata from uploaded documents.
- **Completeness Check:** Identify missing fields, incomplete submissions, and expired documents.
- **Consistency Check:** Cross-verify extracted data within the application and against historical AICTE records.
- **Output:** Generate a structured validation report allowing for applicant rectification if permitted.

4.2 Anomaly & Potential Forgery Detection

- **Document Analysis:** Detect suspicious alterations, duplicates, inconsistent metadata, mismatched IDs, and irregular dates.
- **Explainable AI:** Each flag must explicitly state the reason and link to the specific evidence (e.g., "Mismatched font detected in date field").
- **Status:** Treat findings as preliminary indicators requiring human verification.

4.3 Verification Through Original Issuing Authorities

- **Integration Mechanisms:** Support APIs, official portals, digitally signed records, QR codes, and Unique Reference Numbers (URNs).
- **Verification Workflow:** Compare submitted details against the authorized source.
- **Status Logging:** Record the source, timestamp, and result (Verified, Mismatch, Expired, Pending, Source Unavailable, Manual Review Required).
- **Fallback:** Route to manual confirmation if automated verification fails or is inconclusive.

4.4 Regulatory & Infrastructure Compliance Assistance

- **Configurable Checklists:** Rules derived from the AICTE Approval Process Handbook.
- **Evidence Review:** Assist in analyzing photos, videos, floor plans, and certificates.
- **Computer Vision (Optional/Feasible):** Flag visible discrepancies in infrastructure evidence.
- **Advisory Status:** All automated infrastructure findings are advisory and subject to official review.

4.5 Validation Dashboard, Alerts, & Audit Trail

- **Centralized Dashboard:** Display application-wise status, deficiencies, verification outcomes, and pending manual actions.
- **Automated Alerts:** Notify officials of missing data, suspected anomalies, and pending verifications.
- **Audit Trail:** Maintain a strict, immutable log of rules applied, records checked, API responses, alerts generated, observations, and actions taken.
- **Security:** Role-based access control (RBAC), secure document handling, and data retention safeguards.

5. Non-Functional Requirements (Safeguards)

- **Configurability:** All checks must be adjustable to reflect updates in the AICTE Approval Process Handbook.
- **Security & Privacy:** Authorized, role-based access only. No bypassing of external database controls. Protection of institutional and personal data.
- **Explainability:** AI-generated flags must be supported by evidence.
- **Human-in-the-Loop:** The system provides decision-support only. Statutory scrutiny and final decisions remain with competent authorities.

6. Deliverables

1. Functional prototype (designed for integration with AICTE portal).
2. Configurable regulatory validation rules engine.
3. OCR and document extraction module.
4. Anomaly detection and cross-document consistency checker.
5. Document verification workflow (integrating with external sources).
6. Compliance assistance module (documentary & infrastructure).
7. Admin/Reviewer Dashboard with audit history.

7. Success Metrics (Evaluation Criteria)

- **Accuracy:** Document extraction accuracy, validation accuracy, and reliable anomaly detection.
- **Efficiency:** Average verification time and reduction in manual processing effort.
- **Integration Success:** Successful source-verification rate with external APIs.
- **Reliability:** Timeliness of alerts and reliability of discrepancy identification.

8. Technology Stack Suggestions

Component	Recommended Technology	Purpose / Notes
Backend	Python (FastAPI)	Asynchronous handling of heavy document processing workloads.
AI / LLM	Google Gemini	Structured JSON schemas for data extraction and anomaly flagging.
Document Processing	PyMuPDF / pypdf	Extraction of raw text, metadata, and structural layout.
Database	PostgreSQL	Robust relational data handling and immutable audit trails.
Frontend	React.js / TypeScript	Interactive dashboard for reviewers and system admins.
Computer Vision	OpenCV / TensorFlow	Optional custom visual checks on infrastructure evidence.